

Internship Machine Learning



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Keywords. Energy-based model, discrete space, noise-contrastive estimation.

Targeted departments. Electrical & Computer Eng., Computer Science, Statistics.

Expected outputs. Machine Learning conference submission.

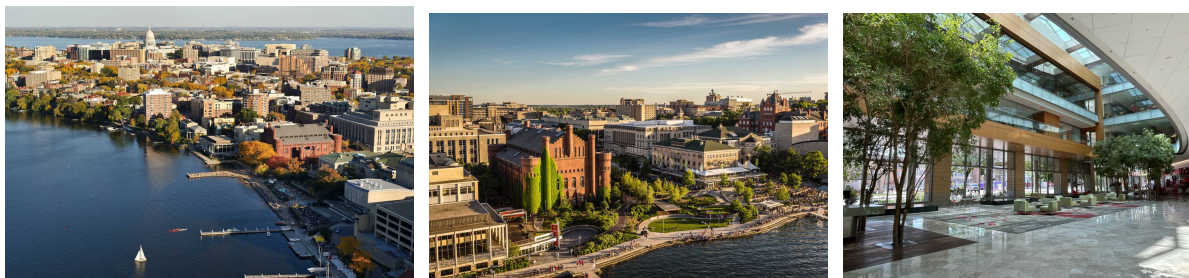
Context. A standard goal in statistics and machine learning is to learn the probability distribution from finite data samples. A long-standing effort has been to parameterize the probability model using neural networks, because they are expressive. One such parameterization is called energy-based models. Learning such models is hard and has led to the development of many famous methods over the past two decades: noise-contrastive estimation, score matching, contrastive divergence. Most methods would not scale to image datasets such as ImageNet, but that has changed in the last year or two [[Guth et al., 2025](#), [Yu et al., 2026](#)]. There is finally a very timely opportunity to make energy-based modelling scalable and to understand what has made this finally possible, after two decades of research.

Goals. The goal of this project is to extend previous work by our group to discrete space. A precise roadmap is ready and has been started, and this project is expected to lead to a publication. At a high level, the project follows how score matching, as a way for estimate energy-based models, was generalized from a continuous space [[Hyvärinen, 2005](#)] to a discrete space [[Meng et al., 2022](#)]. There will be two components to this project:

- **Mathematical.** The candidate will write out the mathematical links between energy-based modelling in continuous and discrete spaces. This is very closely linked with so-called “discrete diffusion models”.
- **Coding.** The candidate will deploy the new algorithm to images, text, and other discrete modalities.

Expected outcomes. With sustained work, this is expected to lead to a machine learning conference submission.

Environment. The position is based in my [research group](#) at the University of Wisconsin–Madison. We work on machine learning, with a particular focus on generative models. Our research combines mathematical foundations, practical algorithm development, and applications in areas such as brain imaging and molecular science.



Requirements. We are seeking a highly motivated master’s or early-stage PhD student with strong programming skills and a mathematical background. Experience in computer vision and a previous publication in top-tier machine learning conferences (*e.g.* NeurIPS, ICML, ICLR, AISTATS, CVPR) is appreciated but not required.

How to apply. Please email ochehab@wisc.edu with the subject line [Internship] Discrete EBM and attach:

1. a CV named `firstname_lastname_CV.pdf`, including a link to your Google Scholar profile, if applicable;
2. a short statement named `firstname_lastname_cover.pdf`, containing one paragraph explaining your interest in the position and what you would bring to the project.

References

Florentin Guth, Zahra Kadkhodaie, and Eero P Simoncelli. Learning normalized image densities via dual score matching. In *Conference on Neural Information Processing Systems*, 2025.

Hanlin Yu, RuiKang OuYang, Partha Kaushik, Arto Klami, Michael U. Gutmann, and Omar Chehab. Learning energy-based models from stochastic interpolants using spatiotemporal differences, 2026.

Aapo Hyvärinen. Estimation of non-normalized statistical models by score matching. *Journal of Machine Learning Research*, 6(24):695–709, 2005.

Chenlin Meng, Kristy Choi, Jiaming Song, and Stefano Ermon. Concrete score matching: Generalized score matching for discrete data. In *Advances in Neural Information Processing Systems*, volume 35, pages 34532–34545. Curran Associates, Inc., 2022. doi: 10.52202/068431-2502.